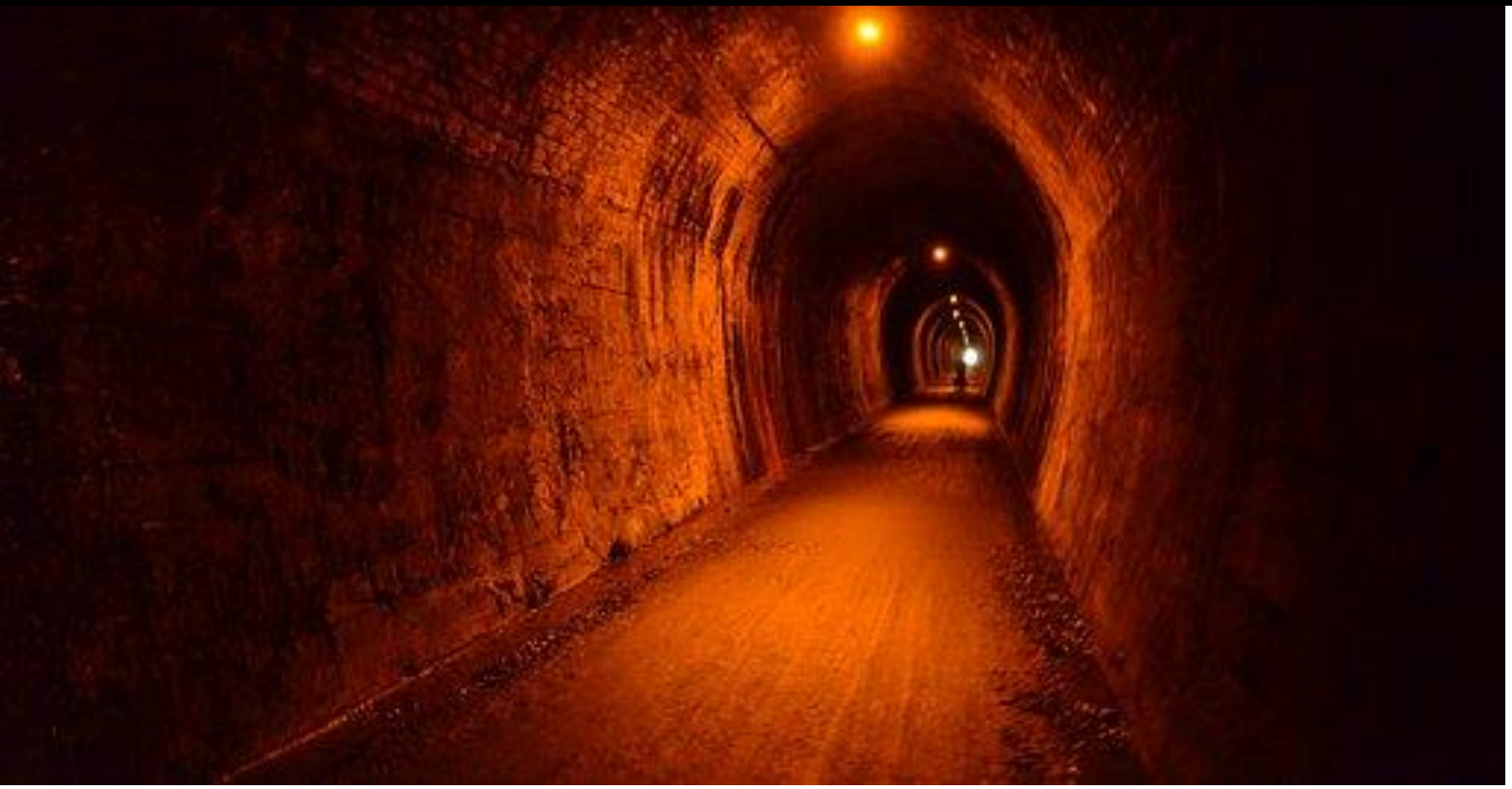
EART11200: Natural Scientist's Toolkit



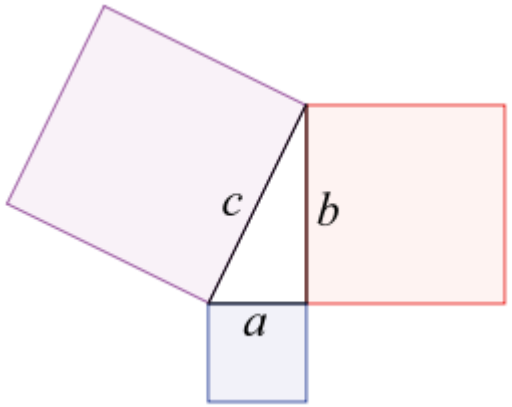


This week considers a geological problem: gold mining
The basic tools cover Pythagoras theorem and trigonometry.

Maths knowledge required

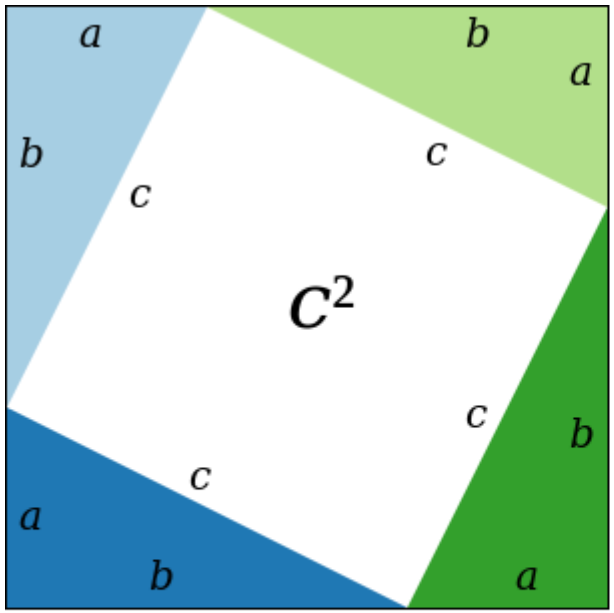
- Pythagorean theorem
- Radians and degrees
- Trig ratios.
- *These concepts come up in many, many, many areas of science. They are used in simple problems and difficult problems too, because they are so fundamental!*

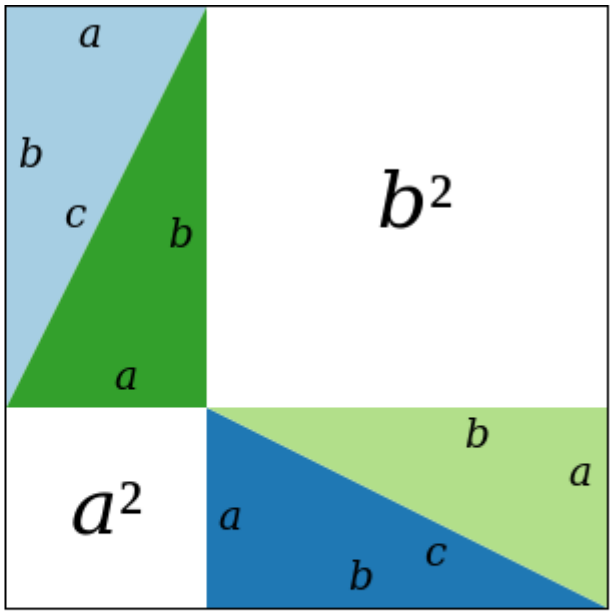
Pythagorean theorem



For a right-angled triangle:

"The square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides"





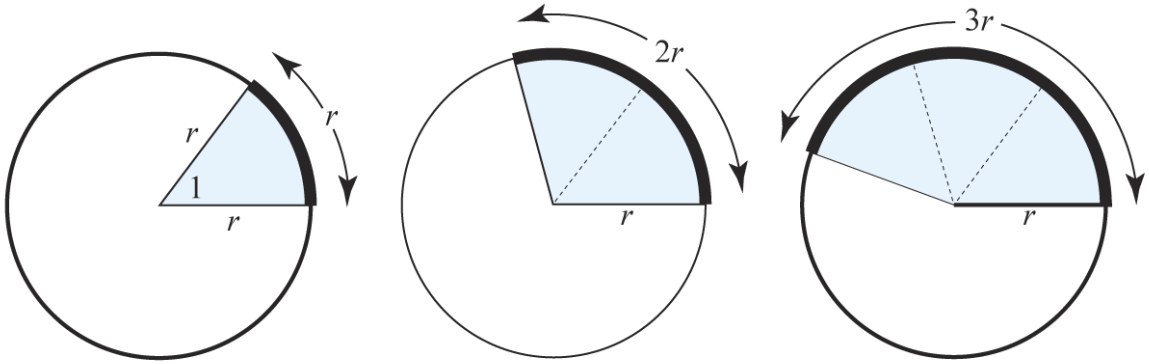
$$c^2 = a^2 + b^2$$

Radians and degrees

Key point

1 radian is the angle subtended at the centre of a circle by an arc length of 1 radius.

Figure 21.5
An arc of length r subtends an angle of 1 radian.



There are 2π radians at the centre of a full circle, and 360°

NB when doing calculations involving angles make sure your calculator is set to the unit that matches your input (radians or degrees).

Key point $\pi = 180°$

WORKED EXAMPLES

21.1 Convert to degrees

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) 0.7π (e) 1 (f) 3 (g) 1.3

Solution

(a) We have

$$\pi = 180°$$

and so

$$\frac{\pi}{2} = \frac{180°}{2} = 90°$$

Hence $\frac{\pi}{2}$ is a right angle.

(b) $\pi = 180°$

$$\frac{\pi}{4} = \frac{180°}{4}$$

$$= 45°$$

Pause the video and try the rest of the examples for yourself, then continue for the answers.

Continued

(c) $\pi = 180^\circ$

$$\frac{\pi}{3} = \frac{180^\circ}{3}$$

$$= 60^\circ$$

(d) $\pi = 180^\circ$

$$0.7\pi = 0.7 \times 180^\circ$$

$$= 126^\circ$$

(e) $\pi = 180^\circ$

$$1 = \frac{180^\circ}{\pi}$$

$$= 57.3^\circ$$

(f) $\pi = 180^\circ$

$$1 = \frac{180^\circ}{\pi}$$

$$3 = 3 \times \frac{180^\circ}{\pi}$$

$$= 171.9^\circ$$

(g) $$1 = \frac{180^\circ}{\pi}$$

$$1.3 = 1.3 \times \frac{180^\circ}{\pi}$$

$$= 74.5^\circ$$

Continued

21.2 Convert to radians

(a) $72°$ (b) $120°$ (c) $12°$ (d) $200°$

Solution

(a) $180° = \pi$

$$1° = \frac{\pi}{180}$$

$$72° = 72 \times \frac{\pi}{180}$$

$$= \frac{2\pi}{5} = 1.26$$

(b) $180° = \pi$

$$1° = \frac{\pi}{180}$$

$$120° = 120 \times \frac{\pi}{180}$$

$$= \frac{2\pi}{3} = 2.09$$

(c) $180° = \pi$

$$1° = \frac{\pi}{180}$$

$$12° = 12 \times \frac{\pi}{180}$$

$$= \frac{\pi}{15} = 0.21$$

(d) $180° = \pi$

$$1° = \frac{\pi}{180}$$

$$200° = 200 \times \frac{\pi}{180}$$

$$= \frac{10\pi}{9} = 3.49$$

Trigonometric ratios

For a right angled triangle:

Key point

$$\sin\theta = \frac{\text{side opposite to } \theta}{\text{hypotenuse}} = \frac{BC}{AC}$$ O/H

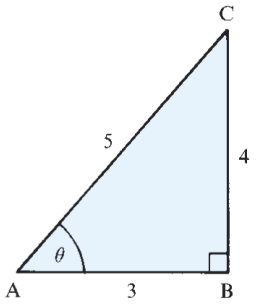
$$\cos\theta = \frac{\text{side adjacent to } \theta}{\text{hypotenuse}} = \frac{AB}{AC}$$ A/H

$$\tan\theta = \frac{\text{side opposite to } \theta}{\text{side adjacent to } \theta} = \frac{BC}{AB}$$ O/A $\tan\theta = \sin\theta / \cos\theta$

WORKED EXAMPLES

22.1 Calculate $\sin\theta$, $\cos\theta$ and $\tan\theta$ for $\triangle ABC$ as shown in Figure 22.2.

Figure 22.2
$\triangle ABC$ for Example 22.1



Solution

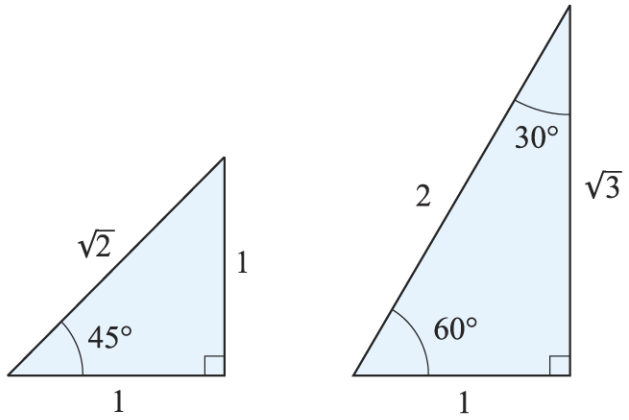
$$\sin\theta = \frac{BC}{AC} = \frac{4}{5} = 0.8$$

$$\cos\theta = \frac{AB}{AC} = \frac{3}{5} = 0.6$$

$$\tan\theta = \frac{BC}{AB} = \frac{4}{3} = 1.3333$$

Figure 22.3

Check for Pythagorean Theorem too!



22.2 Using the triangles shown in Figure 22.3 write down expressions for

(a) $\sin 45°$, $\cos 45°$, $\tan 45°$

(b) $\sin 60°$, $\cos 60°$, $\tan 60°$

(c) $\sin 30°$, $\cos 30°$, $\tan 30°$

Solution

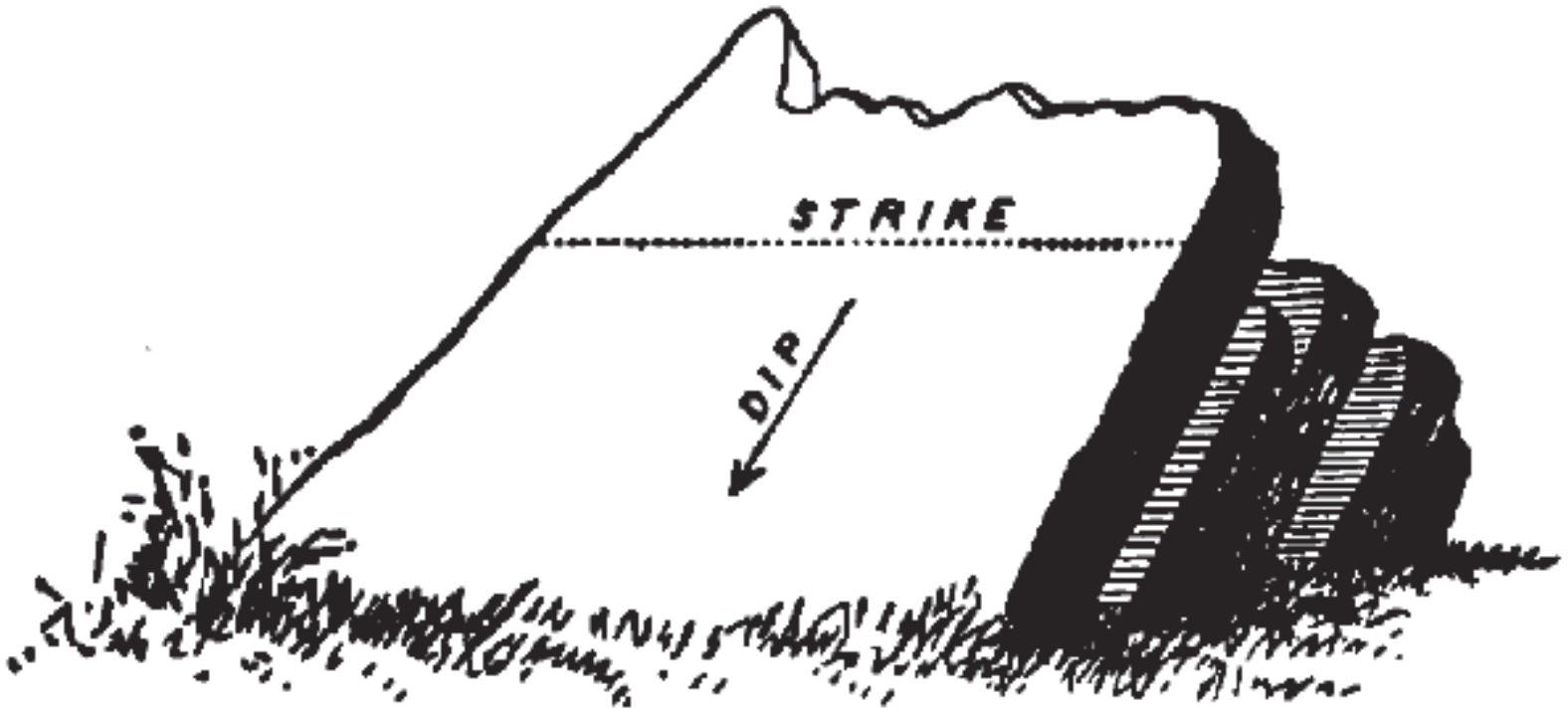
(a) $\sin 45° = \frac{1}{\sqrt{2}}$, $\cos 45° = \frac{1}{\sqrt{2}}$, $\tan 45° = \frac{1}{1} = 1$

(b) $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 60° = \frac{1}{2}$, $\tan 60° = \frac{\sqrt{3}}{1} = \sqrt{3}$

(c) $\sin 30° = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$, $\tan 30° = \frac{1}{\sqrt{3}}$

Notice that we have left our solutions in their exact forms using square roots if necessary rather than given their decimal approximations. These, so called, **surd forms** are frequently used.

Relevance to gold mining?



- What happens beneath the ground?
- How could you find out?